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SUPERSEDING
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MILITARY SPECIFICATION

HOLDER, TELESCOPE MOUNT XM7

1. SCOPE

1.1 This specification covers one type of telescope mount holder, hereinafter referred to as "holder." The holder is primarily intended for use with the M34 Sightunit when firing the Rocket Launcher: XM34 (Little John).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL-P-14232 —Parts, Equipment and Tools for Ordnance Materiel: Packaging of

MIL-I-45208 —Inspection Requirements, General Specification for

MIL-I-45607 —Inspection Equipment, Supply and Maintenance for Ordnance

STANDARDS

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

MUNITIONS COMMAND

F7674705 —Holder, Telescope Mount: XM7

LSQAP 7674705-Supplemental Quality Assurance Provisions List For Holder, Telescope Mount: XM7

IEL 7674705 —Index of Inspection Equipment Lists For Holder, Telscope Mount: XM7

PACKAGING DATA SHEET

MIL-P-14232/P7674705 —Packaging of Holder, Telescope Mount: XM7

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. (see 6.2.)

FSC 1240

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3.2 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Fabrication. The holder shall be manufactured in accordance with Drawing F7674705, "Holder, Telescope Mount: XM7" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.4 General specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.5 Environmental.

3.5.1 Operating temperature. The holder shall meet the applicable requirements of 3.6.7 while exposed to and thermally stabilized, at ambient temperatures of plus 150° and minus 40° Fahrenheit.

3.5.2 Vibration.T beholders shall be vibrated at a constant frequency of 30 cycles per second with an amp]itude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes, plus or minus 15 seconds. Subsequent to vibration the holder shall show no evidence of loose or damaged parts and shall meet all the performance requirements of this specification.

3.6 Performance. With the mounting surface of the bracket of the holder horizontal, the holder shall meet the requirements of 3.6.1 to 3.6.6 inclusive.

3.6.1 Cross level pads. The cross level quadrant pads of the bracket shall be level within SO seconds of arc.

3.6.2 Elevation pads. The elevation quadrant leveling pads of the cross level housing shall be level within 50 seconds of arc.

3.6.3 Indexes.

3.6.3.1 Cross level housing and bracket. When the flat of the dovetail locating surface is perpendicular to the mounting surface of the bracket within 5 minutes of arc, the index line on the cross level housing shall be in coincidence with the index line on the bracket within one-half of the thickness of the index line.

3.6.3.2 Cross level knob and housing. When the flat of the dovetail locating surface is perpendicular to the mounting surface of the bracket within 5 minutes of arc, the adjustable index line on the knob shall be in coincidence with the index on the worm housing.

3.6.4 Cross level travel. Rotation of the cross level knob shall move the cross level housing a minimum of 40 degrees to the left and 40 degrees to the right from the position established when the index line on the worm housing and the cross level housing are in coincidence with their respective indexes.

3.6.5 Backlash. Backlash readings at any position of cross level travel shall not exceed 0.75 roil.

3.6.6 Dovetail position. The angular position of the dovetail shall meet all drawing requirements and the dovetail shall be 45 degrees plus or minus 2 minutes from horizontal.

3.6.7 Operability. The torque required to rotate the cross level knob shall not exceed 12 pound inches when the holder is either elevated or depressed at standard ambient temperature (plus 60° to 90° Fahrenheit), The torque required to rotate the cross level knob shall not exceed the value obtained at standard ambient temperature by more than 3 pound inches when subjected to extreme temperatures of 3.5.1.

4. QUALITY Assurance PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for

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the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Inspection requirements and procedures shall be in accordance with MIL-I-45208.

4.3 Initial production sample. The contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall reinspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Initial production sample failure. Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation. Inspection lot sizes and lot formation shall be in accordance with MIL-STD-105.

4.4.1 Defective units and lots. Defective units and lots shall be processed as specified in MIL-STD-105.

4.5 Material. Control of material shall be in accordance with test data requirements of MIL-I-45208.

4.6 Inspection of components. Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP)-7674705*

4.6.1 Contents of SQAP. The contents Of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of each component covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications, and to reject individual nonconforming items.

4.6.2 Cpmtractors inspection for all components not covered by SQAP. Inspection for these components shall be in accordance with the inspection procedures of the inspection system required by MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the holders.

4.6.3 Control samples. After the initial production sample (4.3) has been accepted, two holders randomly selected from each month's production, or two from each lot of 100, whichever occurs first, shall be submitted to the government to determine compliance with 3.2, 3.3, and 3.5.l. This inspection shall be conducted by the contractor under the surveillance of the government representative at the contractor's facility or government approved laboratory, using test equipment authorized for use on the contract.

4.6.3.1 Control sample failure. Should a control sample fail to meet any one of the requirements indicated in 4.6.3, the contractor shall cease those operations which are causing the rejections and shall make the necessary corrections. VVhen corrections in the process have been made, new control samples randomly selected by the government shall be examined and tested by the contractor under the surveillance of the government. Acceptance of units will be withheld until the resubmitted samples have been approved.

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4.7 Inspection equipment. Inspection equipment shall conform to drawings listed in IEL 7674705, except as provided for under the alternative inspection procedures and inspection equipment requirements of MIL-I-45208. Provisions of MIL-I-45607 shall apply to test equipment provided by the Govermnent or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired, for verification inspection.

4.7.1 Accuracy requirements. Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance except as specified in 4.7.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously. advised in order that necessary appropriate action can be taken..

4.7.2 Inspection equipment design requirements.

, .4.7.2.1 Equipment to perform extreme temperature test. The test equipment shall be in accordance with the: requirements which include pertinent volume and tolerances governing the environmental test conditions and measurements contained in MIL-F-13926 for the operating temperatures specified in 3.5.1.

4.7.2.2 Equipment to perform vibration test. The test equipment shall be of a type of vibrating machine capable of meeting the conditions of 3.5.2. The accuracy of the vibrating machine shall be in accordance with the tolerances contained in MIL-F-13926 for vibration.

4.7.2.3 Equipment to perform operability test. The test equipment shall be commercial type torque wrench with a special adapter to accommodate the cross level knob. The torque wrench type tester shall permit reading with a minimum range from 0 to 15 pound inches within plus or minus 1/4 pound inch throughout the entire zone of excursion in both clockwise and counterclockwise directions.

4.8 Inspection. The classification of defects in table I and II shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection of lots. The AQL specified pertains to each individual characteristic.

4.8.1 Samples selected for inspection of the characteristics listed in table II shall include those samples used for inspection in table I.

4.8.2 Disposition of rejected lots. Rejected lots shall be screened for all defective characteristics. After corrections have been made and the government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in MIL-STD-105.

TABLE I. Classification of defects

Use inspection level II in table III–A with sampling plan table IV–A of Standard MIL–STD–105.

Critical: None defined.

Major: AQL 1.5 percent defective	*Requirement paragraph*	*QA. paragraph*	*Test method*
101. Cross level pads not level	**3.6.1**	**4.9.5**	**Standard Measuring Equipment (SME)**
102. Elevation pads not level	**3.6.2**	**4.9.6**	**Special Test Equipment (STE)-SME**
103. Index line on cross level housing and bracket not in coincidence.	**3.6.3.1**	**4.9.7.1**	**STE-Visual (Vis)**
104. Index line on cross level knob not in coincidence.	**3.6.3.2**	**4.9.7.2**	**STE-Vis**
105. Insufficient cross-level travel	**3.6.4**	**4.9.7.3**	**STE-SME**
106. Backlash excessive	**3.6.5**	**4.9.7.4**	**STE**

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Major: AQL 1.5 percent defective	Requirement paragraph	QA. paragraph	Test method
107. Dovetail position from horizontal not as specified.	3.6.6	4.9.6.5	STE-SME
108. Operability, excessive cross level knob torque.	3.6.7	4.9.7.6	SME
109. Marking missing, incorrect illegible.	3.4	4.9.8	Vis

Minor: None defined.

TAbLE II. Classification of defects

Use inspection level L7 in table III–B with sampling plan table IV–A of Standard MIL–STD–105.

Critical: None defined.

Major: AQL 1.5 percent defective	Requirement paragraph	QA. paragraph	Test method
101. Excessive cross level knob torque at the extreme temperatures	3.5.1, 3.6.7	4.9.2, 4.9.7.6	SME
102. Physical failure post vibration	3.5.2	4.9.3	Vis-Tactile
103. Performance failure post vibration	3.5.2	4.9.3	DST-SME Vis

Minor: None defined.

4.9 Test methods and procedures.

4.9.1 Fabrication. To determine compliance with 3.3, the holder shall be visually inspected for proper fabrication in accordance with the specified drawings, and drawings pertaining thereto. Dimensional checks shall be made by use of standard measuring equipment. This inspection shall be performed on the initial production and control samples (4.3 and 4.6.3).

4.9.2 Operating temperature. The operating temperature test consists of exposing the holder to the temperatures indicated in 3.5.1 by means of equipment complying with the requirements of 4.7.2.1. The test shall be conducted in accordance with Procedure I of MIL-F-13926, and the conditions of 3.5.1 shall apply. While at these temperatures, the test for the requirements of 3.6.7 shall be made by use of equipment complying with the requirements of 4.7.2.3.

4.9.3 Vibration. The vibration test shall be performed by means of a vibrating machine compying with the requirements of 4.7.2.2. The holder shall be rigidly mounted on the vibrating machine, and vibrated in accordance with the requirements indicated in 3.5.2. Subsequent to vibration, the holder shall be subjected to a visual and tactile examination for evidence of loose or damaged parts, after which the holder shall be subjected to and meet all performance requirements of this specification.

4.9.4 Performance. The performance requirement test shall be accomplished with the holder located on a cross leveled surface plate which shall position the holder as prescribed in 3.6, in conjunction with a special testing device. The device shall be so constructed as to be positioned within the dovetail of the holder located as required herein, and shall include sufficient surfaces and mediums to test for the requirements of 3.6.1,3 .6.2,3.6.3,3.6.4 and 3.6.6. The surface of the device, used to support the medium which will establish perpendicularity of the dovetail flat on the holder shall be ninety degrees plus or minus ten seconds to the flat of the dovetail surface on the device. This surface shall also be forty five degrees plus or minus ten seconds to the axis of the dovetail. Drawings depicting testing devices are listed on IEL 7674705.

4.9.5 Cross level pads. With the mounting surface of the bracket of the holder in the horizontal position, a ten second bench level shall be positioned across the top of the pads. Observe the indication on the bench level to determine conformance to the requirements specified in 3.6.1.

4.9.6 Elevation pads, With the holder posi-

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tioned as in 4.9.5 the test shall be accomplished with a special testing device inserted in the dovetail of the holder. With a ten second bench level across the top surface of the special testing device (perpendicularity gage), level the top surface of the perpendicularity gage by rotating the cross level knob until the bench level vial bubble is central. Locate a ten second bench level across the tops of the elevation pads; observe the indication on the bench level to determine conformance to the requirements specified in 3.6.2.

4.9.7 Indexes.

4.9.7.1 Cross level housing and bracket. Subsequent to the above test, the unit under test shall be visually examined to determine conformance to the requirements specified in 3.6.3.1.

4.9.7.2 Cross level knob and housing. Subsequent to the above examination, the unit under test shall be visually examined to determine conformance to the requirements specified in 3.6.3.2.

4.9.7.3 Cross kwet travel. This test shall be accomplished using an optical protractor in place of the bench level with the above alinement set up. With the index line on the worm housing and the cross level housing in coincidence with their respective indexes, the protractor shall be utilized to determine conformance to the requirements of 3.6.4 when the cross level housing is rotated to its limits of excursion in each direction specified.

4.9.7.4 Backlash. Backlash shall be checked by means of a special testing device. The device shall be graduated in mil increments with a means of being clamped over the index marker of the cross level knob. The test shall be accomplished utilizing the same set up as in 4.9.4 with the device to check perpendicularity retained in the unit under test, except that the precision bench level shall be replaced with an optical protractor, With the protractor set at the angle to be tested, and placed on the special testing device which fits into the dovetail of the holder, the cross level knob shall be rotated by a clockwise movement and then by a counterclockwise movement until the level bubble of the protractor reads zero at each movement. The difference in readings of each movement, as taken on the cross level knob, is considered backlash. The difference in readings of each directional movement shall not exceed the requirements specified in 3.6.5.

4.9.7.5 Dovetail position. The test shall be accomplished using the same set up as in 4.9.4, except that an optical protractor shall be used across the top surface of the special testing device to determine conformance to the requirements specified in 3.6.6.

4.9.7.6 Operability. Operability shall consist of measuring the torque required to rotate the cross level knob of the holder through full travel at standard ambient temperature by means of a torque wrench with an adapter capable of accepting the cross level knob. These torque values shall then be compared with the operating temperature test of 3.5.1, and the difference shall not exceed the prescribed value in 3.6.7.

4.9.8 Part identification and marking. Marking for identification shall be visually inspected for conformance with MWF-13926.

4.9.9 Preservation, packaging, packing and marking. The quality assurance provisions for the preservation, packaging, packing and marking shall be in accordance with MI~P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P7674705. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Holder, Telescope Mount: XM7 is a light weight holder primarily intended for use with rocket launchers as a cross leveling and mounting device for sightunits of the M34 series for both direct and indirect fire.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number
(d) Provisions for initial production sample including date of delivery (see 3.1).

6.3 Design information. Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.4 Definitions

6.4.1 Standard measuring equipment (SME). Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.4.2 Special test equipment (STE). Special test equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.4.3 Tactile. Skillful manipulation or sensing.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto,

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